

■ PRD Generation and Critique – CourseLLM (Team Submission)

1. Summary of LLM Feedback

After completing our first draft of the CourseLLM – AI Course Companion PRD, we used ChatGPT to provide a structured critique. The LLM offered detailed and constructive feedback, focusing on clarity, prioritization, and alignment between our vision and the concrete features we defined.

Key feedback points included:

- **Vision and Objectives:** The overall vision was strong, but the LLM encouraged us to emphasize *how* CourseLLM promotes deeper learning rather than only describing its functionality.
- **Scope and Focus:** It advised us to sharpen the boundaries between the MVP and future enhancements, keeping the initial release realistic.
- **User Stories:** The model pointed out that our user stories were inconsistent in format and suggested converting them all into the clear 'As a [user], I want [goal], so that [benefit]' template.
- **Pedagogical Connection:** It recommended explicitly linking each feature to measurable learning outcomes and to the Socratic guidance method described in our objectives.
- **Structure and Consistency:** We were advised to maintain a consistent structure (Vision → Features → Stories → Acceptance Criteria → Metrics) and separate the teacher and student perspectives more clearly.

2. Revisions We Implemented

Following the feedback, we made several improvements to our PRD:

- Rewrote the **vision** section to highlight learning reinforcement and reflection as the main pedagogical goals.
- Added **quantifiable success indicators** (e.g., completion rates, reasoning depth, teacher dashboard insights).
- Reformatted all **user stories** to match a consistent structure and linked them directly to the relevant requirements.
- Expanded the **acceptance criteria** for core features such as guided chat, resource filtering, and student progress tracking.
- Split the PRD into two clear perspectives — one for **students** and one for **teachers/TAs** — to better reflect the system's dual purpose.
- Improved the **data model** section to represent conversation history and feedback logs more accurately.

3. Reflection on the Process

Working as a team with the LLM felt like having a collaborative reviewer that combined a teaching assistant's structure with a product manager's attention to detail. It identified issues we had overlooked — especially around clarity, educational focus, and scope control — and helped us express our ideas more coherently.

The process also sparked valuable internal discussion: we debated which of its suggestions fit our course context, which led to a deeper shared understanding of requirements engineering principles. We realized that while LLMs can't replace human judgment, they significantly accelerate the *iteration* phase of writing and reviewing PRDs.

Overall, using ChatGPT as a critique partner improved both the quality of our document and our teamwork process. It made us more deliberate about defining measurable outcomes and maintaining alignment between the product vision and the actual system behavior.

■ Summary

- **Team tool used:** ChatGPT (GPT-5)
- **Main improvements:** Clearer structure, measurable outcomes, refined user stories, and better persona separation.
- **Team takeaway:** LLMs can serve as intelligent reviewers that strengthen group reasoning, promote collaboration, and improve documentation quality.

■ LLM Critique Conversation

Full critique available at:

<https://chatgpt.com/share/690fb791-1dfc-8012-976c-03fb7e71d9dd>